

Name _____

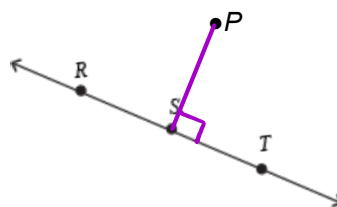
Accel Geo – Unit 2 Section 3-4 – Proofs with Perpendicular Lines

in mm

Measure the distance from point P to each labeled point on the line.

$$RP = 20 \text{ mm} \quad SP = 16 \text{ mm}$$

$$TP = 22 \text{ mm}$$

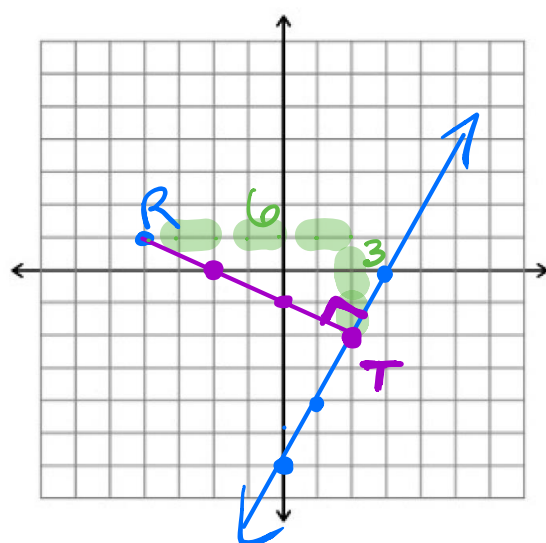


The shortest distance between a point and a line is the perpendicular segment.

To find the distance, use the distance formula.

Example:

Graph the line $y = 2x - 6$ and $R(-4, 1)$



$$\sqrt{6^2 + 3^2} \quad \sqrt{45}$$

$$\sqrt{36 + 9}$$

Step 1: What is the slope of the given line?

$$m = \frac{2}{1} \quad \frac{\text{rise}}{\text{run}}$$

Step 2: What is the slope of a line perpendicular to the given line?

$$m = -\frac{1}{2}$$

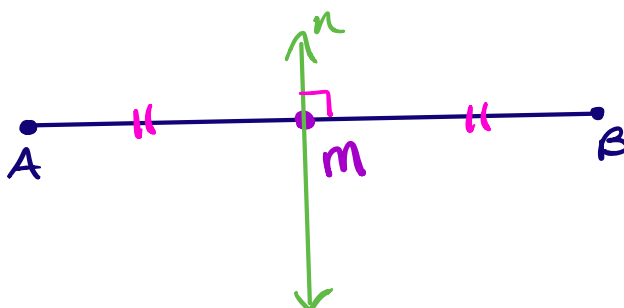
Step 3: From the given point, use the slope from step 2 to find the point of intersection with the given line. $T(2, -2)$

Step 4: Use the distance formula to find the length of the perpendicular segment.

$$RT \approx 6.71 \text{ un.}$$

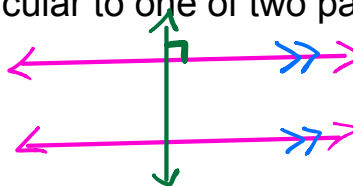
Perpendicular Bisector of a segment is the line that passes through the midpoint of the segment at a Right or 90° angle.

Sketch: line $n \perp \overline{AB}$ and line n bisects $\overline{AB}$ at M .

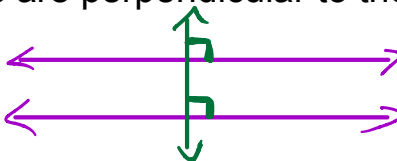


More Theorems

1. In a plane, if a transversal is perpendicular to one of two parallel lines, then it is perpendicular to the other line.



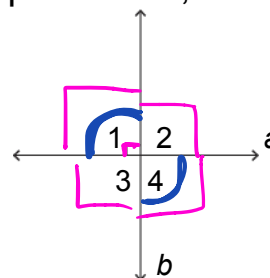
2. In a plane, if two lines are perpendicular to the same line, then they are parallel to each other.



Prove the statement: If two intersecting lines are perpendicular, then they intersect to form four right angles.

Given: $a \perp b$

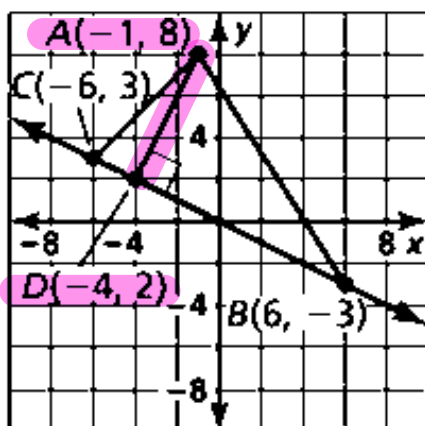
Prove: $\angle 1, \angle 2, \angle 3,$ and $\angle 4$ are right angles



Statements	Reasons
① $a \perp b$	① given
② $\angle 1$ is a right angle	② Def of Perpendicular Lines
③ $\angle 1 \cong \angle 4$	③ Vertical $\angle \cong$ Thm (from pic)
④ $m\angle 1 = 90^\circ$	④ Def of right angle
⑤ $m\angle 4 = 90^\circ$	⑤ Substitution Prop.
⑥ $\angle 1$ and $\angle 2$ form a linear pair	⑥ Def. of Linear Pair (from pic)
⑦ $\angle 1$ and $\angle 2$ are supplementary	⑦ Linear Pair Post.
⑧ $m\angle 1 + m\angle 2 = 180^\circ$	⑧ Def. of Supplementary
⑨ $90^\circ + m\angle 2 = 180^\circ$	⑨ Substitution Prop.
⑩ $m\angle 2 = 90^\circ$	⑩ Subtraction Prop.
⑪ $\angle 2 \cong \angle 3$	⑪ Vert. $\angle \cong$ Thm
⑫ $m\angle 3 = 90^\circ$	⑫ Substitution Prop
⑬ $\angle 1, \angle 2, \angle 3,$ and $\angle 4$ are right angles	⑬ Def of Right Angles

More Examples for Section 3-4

1. Find the distance from point A to $\overline{BC}$.



$$AD = \sqrt{(8-2)^2 + (-1-4)^2}$$

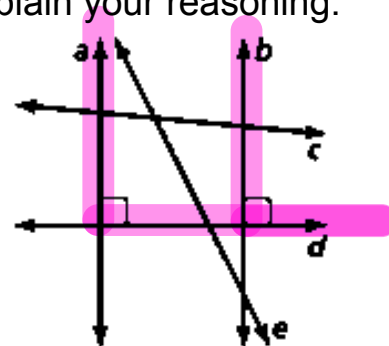
$$AD = \sqrt{45}$$

$$AD \approx 6.71 \text{ units}$$

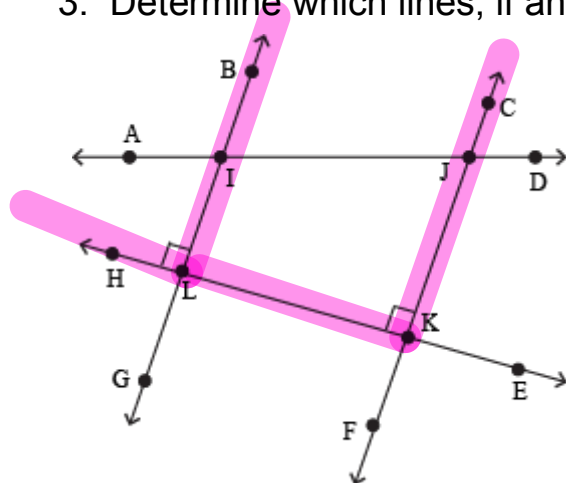
2. The diagram shows the layout of walking paths in a town park. Determine which lines, if any, must be parallel in the diagram. Explain your reasoning.

line a $\parallel$ line b

If 2 lines are perpendicular to the same line, then they are parallel to each other.



3. Determine which lines, if any, must be parallel in the diagram. Explain.



$\overleftrightarrow{BG} \parallel \overleftrightarrow{CF}$

If 2 lines are perpendicular to the same line, then the lines are parallel to each other.